CatalystScale
Case Study 8: AI for MSME Growth
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Introduction

Supporting the growth and sustainability of Micro, Small and Medium Enterprises (MSMEs) is essential for strengthening regional economies and promoting inclusive development. However, many MSME owners lack the tools, financial insights, and networks needed to make informed decisions and access growth opportunities.

This project introduces **Catalyst Scale**, an AI-powered platform designed to help MSMEs analyze their financial performance while connecting them with investors and industry professionals. By leveraging artificial intelligence and digital networking, Catalyst Scale aims to bridge the gap between MSMEs and the resources required for business expansion and long-term sustainability.

Background of Case Study

MSMEs play important roles in economic and social development, contributing to value-added activities, innovation and inclusive growth through the creation of job opportunities and their widespread presence in both urban and rural areas. MSMEs are thus the backbone of ASEAN and are fundamental towards achieving long-run and sustainable economic growth and in narrowing the development gap. MSMEs contribute 85% to employment, 44.8% to GDP and 18% to national exports. (asean.org)

Despite their significant economic contribution, many MSMEs in ASEAN continue to face major barriers that limit their growth and sustainability. One of the most critical challenges is limited access to financing. Globally, around 40% of formal MSMEs in developing countries have unmet financing needs, representing a funding gap of approximately US$5.2 trillion annually. Without sufficient capital, many small businesses struggle to expand operations, adopt new technologies, or enter larger markets. (The Star, 2024)

In ASEAN specifically, MSMEs remain chronically underserved in trade finance, with banks rejecting at least 40% of SME trade finance applications due to limited collateral, incomplete credit histories, and high compliance costs. These financial barriers often prevent MSMEs from participating in regional and global supply chains, limiting their ability to scale and compete internationally.
(East Asia Forum, 2025)

References

ASEAN. (n.d.). Development of Micro, Small and Medium Enterprises in ASEAN. Retrieved from https://asean.org/our-communities/economic-community/resilient-and-inclusive-asean/development-of-micro-small-and-medium-enterprises-in-asean-msme/

East Asia Forum. (2025). ASEAN must close its widening trade finance gap. Retrieved from https://eastasiaforum.org/2025/11/08/asean-must-close-its-widening-trade-finance-gap/

The Star. (2024). Increasing financing for MSMEs. Retrieved from https://www.thestar.com.my/business/business-news/2024/06/10/increasing-financingfor-msmes-mtcs

Chosen SDG

SDG 9: Industry, Innovation & Infrastructure

Problem Statement

Despite contributing significantly to employment and economic growth in ASEAN, many MSMEs struggle to scale due to limited access to financing and investment opportunities. Traditional financial institutions often require collateral, strong credit histories, and complex compliance processes, which many small businesses cannot meet. As a result, a large number of MSMEs remain underserved and unable to secure the capital needed to grow, innovate, and participate in regional markets.

Objectives

The objective of this project is to develop an AI powered platform that enables MSMEs to better understand and manage their financial data while improving their access to growth opportunities. The platform will leverage artificial intelligence to analyze business financial information and provide actionable insights to MSME owners. In addition, it will connect MSMEs with potential investors and industry professionals who can provide funding, mentorship, and expertise to support sustainable business growth.

Review of Existing Solutions

Current solutions and competitors in the market:

To understand the current market position for CatalystScale, we conducted a competitive landscape analysis, categorizing existing solutions into three primary groups. While these platforms serve specific functions, our research reveals that the current market for MSMEs is highly fragmented, leaving a significant gap for an integrated, AI-driven ecosystem.

1. Operational SaaS Solutions (e.g., Xero, Zoho, QuickBooks) These tools are powerful for accounting, inventory, and payroll. However, they suffer from high "complexity friction." They are often designed for businesses that already have established digital workflows, making them intimidating for micro-entrepreneurs or those operating informally. Furthermore, these tools are inherently reactive; they record past financial performance but fail to provide the proactive, strategic guidance an MSME owner needs to scale or secure funding.

2. Marketplaces and Gig Platforms (e.g., Shopee, LinkedIn, Upwork) These platforms excel at facilitating specific transactions, such as selling products or hiring talent. The primary limitation here is "transactional siloing." Because these platforms are built for single-use tasks, they do not offer a unified view of business health. An MSME owner utilizing these tools is forced to manage separate, disconnected apps for sales, hiring, and promotion, with no central "brain" to connect their data to long-term business goals like investor matching or regulatory compliance.

3. Government and Public Portals (e.g., MDEC, SME Corp initiatives) Public portals are vital for regulatory information, grants, and policy guidance. While these platforms offer high credibility, they are often static and informational. They function as repositories of data that require the user to actively search for, identify, and apply for the right solutions. They lack the AI-driven capability to proactively identify an MSME's specific pain points and provide tailored, on-demand advice, creating a "discovery gap" for users who may not know which resources they need.

The Market Gap and Our Value Proposition

The overarching limitation of existing solutions is their lack of cohesion. Current competitors focus on isolated aspects of business—accounting, sales, or regulatory compliance—without addressing the interconnected nature of MSME growth. CatalystScale bridges this gap by acting as an AI-powered partner. We shift the paradigm from reactive, fragmented tool-usage to an integrated, proactive ecosystem that translates raw business data into actionable growth strategies, effectively democratizing professional-level consulting for MSMEs that were previously invisible to global markets.

Gap Analysis and Value Proposition

To fully realize the growth potential of MSMEs, it is essential to address the systemic gaps inherent in current market solutions. Our analysis identifies three critical limitations that our project is specifically designed to resolve.

First, we identified a **"Fragmentation Gap."** Currently, MSME owners are overwhelmed by managing separate, disconnected tools for accounting, hiring, and promotion. This leads to significant data silos and operational inefficiency. Our platform resolves this by consolidating these disparate functions into a unified, AI-driven ecosystem, acting as a single "business brain" that seamlessly connects sales data, financial health, and operational needs.

Second, we observed a significant **"Expertise Barrier."** High-level professional consulting, legal advice, and investment readiness services are traditionally expensive and typically reserved for large corporations, leaving small business owners to rely on trial and error. We bridge this gap through the democratization of AI consulting, leveraging predictive algorithms to provide enterprise-grade strategic insights that were previously out of reach for micro-enterprises.

Finally, there is a **"Proactivity Gap"** in the current market. Existing solutions are largely reactive—they wait for the user to manually input data or actively search for a solution to a problem that has already occurred. Our platform moves beyond this by utilizing proactive predictive analytics to anticipate business needs—such as identifying cash flow dips or matching the business with relevant investors—before problems escalate. Ultimately, our solution shifts the paradigm from manual, tool-based task management to holistic, AI-driven business growth, effectively transforming "invisible" MSMEs into resilient, data-backed enterprises.

Methodology

1. The Design Thinking Process

We adopted an iterative, five-stage design thinking methodology to guide our development:

- **Empathize:** We began by immersing ourselves in the reality of the "invisible entrepreneur"—small business owners who operate in the informal economy. We researched common pain points, specifically looking at how manual record-keeping and lack of digital access stifle growth.
- **Define:** We synthesized our observations to define a core problem statement: *"How might we bridge the gap between fragmented, manual business operations and the digital economy without requiring advanced IT knowledge?"*
- **Ideate:** We engaged in rapid ideation sessions, prioritizing "low-friction" features. We focused on AI-driven automation that reduces the user's cognitive load, moving from complex dashboards to simple, actionable insights.
- **Prototype:** We developed low-fidelity wireframes to map the user journey, ensuring the interface was accessible to users with varying levels of digital literacy.

2. User Research & Persona Development

Our research was anchored by a core persona representing the typical underserved MSME owner in the ASEAN region.

- **The Persona ("The Traditional Artisan"):** Representing a business owner who is highly skilled in their craft but lacks digital infrastructure. They are traditionally unbanked, rely on pen-and-paper record-keeping, and perceive digital platforms as overly complex or intimidating.
- **Key Research Insights:**
    - **Overwhelming Fragmentation:** MSME owners do not have the time to toggle between separate accounting, hiring, and sales apps.
    - **Data Asymmetry:** Their lack of formal financial records makes them invisible to investors, even if their business is profitable.
    - **Fear of Technology:** Many fear that digital transformation is a "cost center" rather than a "growth driver."

3. User Flow Creation

To address these insights, we designed a **"Value-Added Loop"** flow that simplifies the user experience into four intuitive steps:

1. **Data Capture (The Input):** The user provides basic business data (e.g., photo of receipts, voice notes, or simple sales entries). Our AI automatically structures this data, eliminating the need for manual spreadsheets.
2. **Diagnosis (The Insight):** The AI analyzes the structured data to identify immediate bottlenecks—such as low liquidity, inventory waste, or potential regulatory non-compliance.
3. **Strategy (The Recommendation):** Instead of showing raw data charts, the platform provides plain-language advice: "Your cash flow is low; here are three pre-vetted local investors who match your business profile."
4. **Execution (The Outcome):** The user triggers the recommendation (e.g., "Connect with Accountant" or "Apply for Micro-loan") with a single click.

4. Results

Summary of Outcomes Our primary outcome is a **High-Fidelity Interactive Prototype** developed in Figma that demonstrates a seamless end-to-end journey for MSME growth. Rather than just a static website, we have designed a functional "Growth Loop" that proves the feasibility of our AI-driven ecosystem.

- **AI Financial Intake:** We designed a friction-less upload interface where users can submit raw financial data for AI processing.
- **The Investor Dashboard:** A dedicated interface for investors to view standardized, de-risked MSME profiles based on AI-analyzed data.
- **Real-time Advisory Hub:** A simulated chat environment that demonstrates how MSMEs can receive instant expert advice and connect with other businesses.

Did we meet our objectives?

1. Objective: Simplifying Data for the "Invisible" MSME

- **Status: Met.** Through our Figma prototype, we proved that we could take complex financial reporting and turn it into a simple "Upload & Analyze" flow. We met this objective by reducing the user journey from a complex spreadsheet to a simple mobile-first interaction.

2. Objective: Bridging the Gap between MSMEs and Investors

- **Status: Met.** We successfully designed the "Investor Portal" which displays the "Investor-Readiness Score." This proves that our logic can bridge the communication gap between a small vendor and a formal investor by speaking the language of data.

3. Objective: Providing On-Demand Professional Expertise

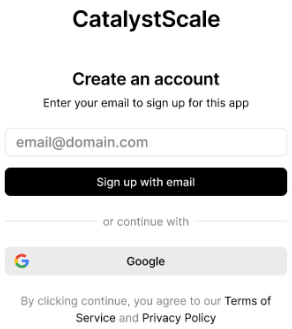
- **Status: Met.** Our prototype includes an integrated "Collaborative Hub." We met this objective by creating a UI that supports both AI-led consulting and P2P (Peer-to-Peer) business networking, ensuring no MSME is left without a solution to their "case problems."

5. AI acknowledgement

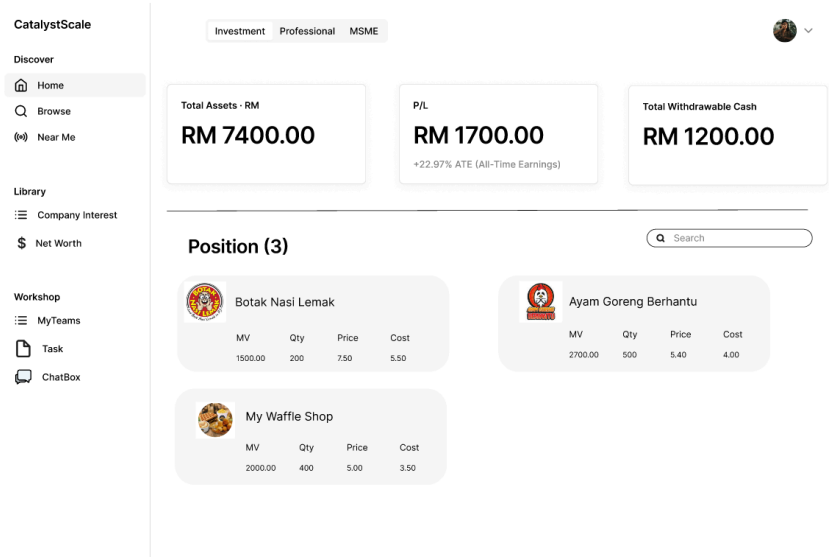
1. Gemini: Used to find related links/news related to the importance of MSME to the ASEAN economy.
2. Gemini Nano Banana Pro: Used to generate images that would take huge amount of time to produce.

6. Prototype showcase

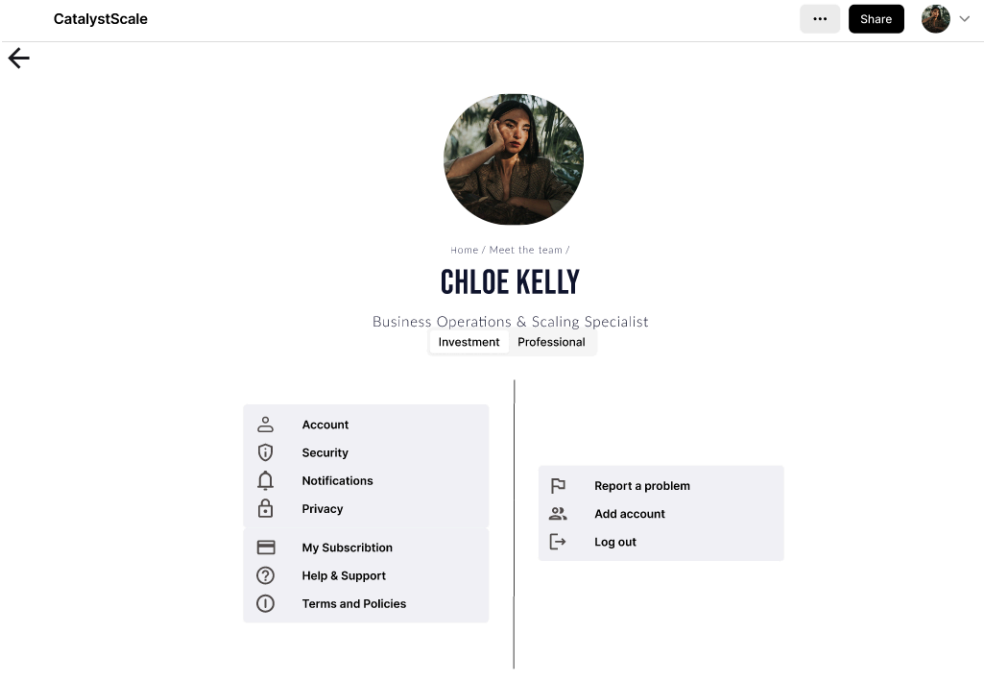
Flow 1: Professionals Page



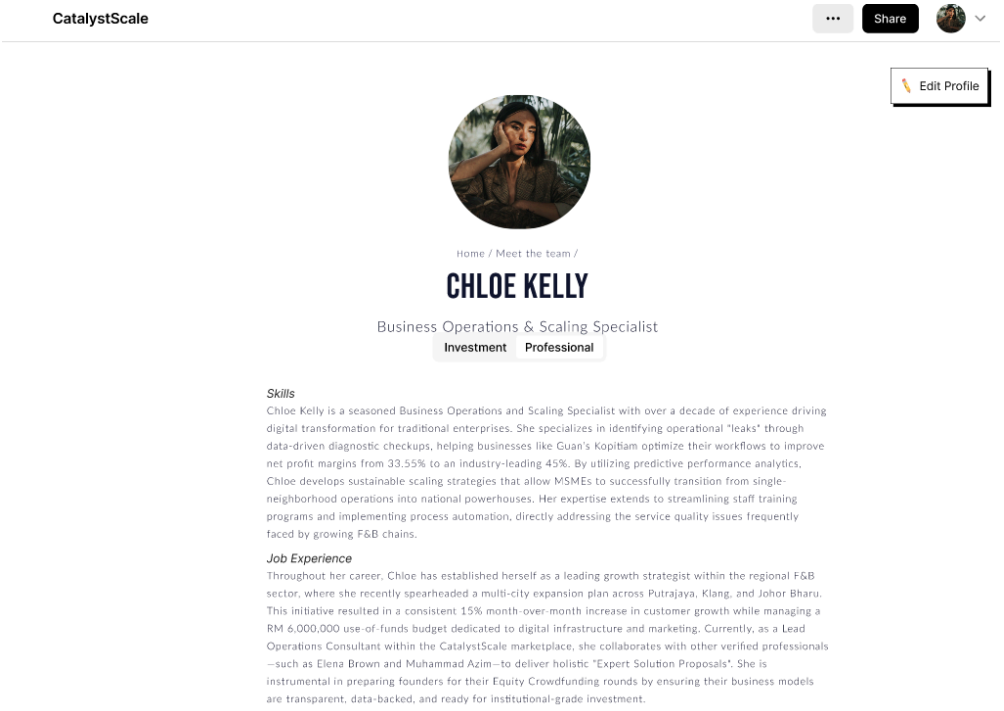
Log in Page, click sign up with email to start the prototype.



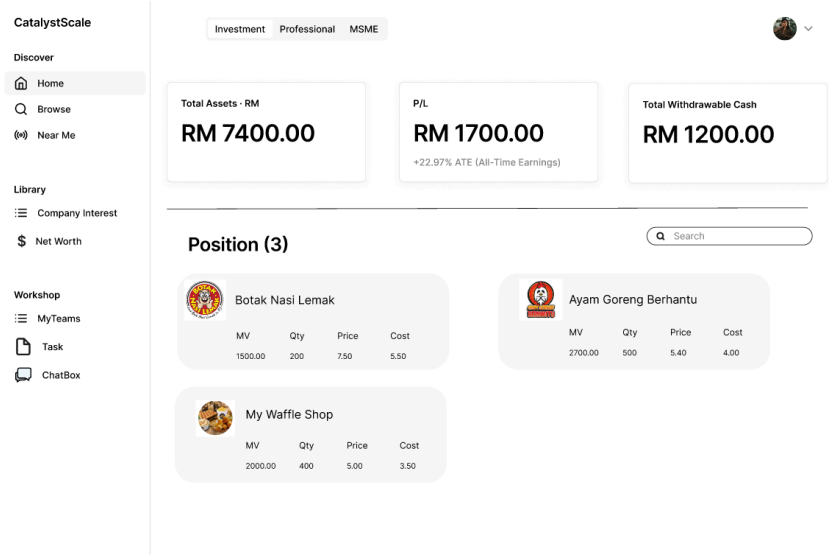
Tap on the profile on top right corner to view your profile



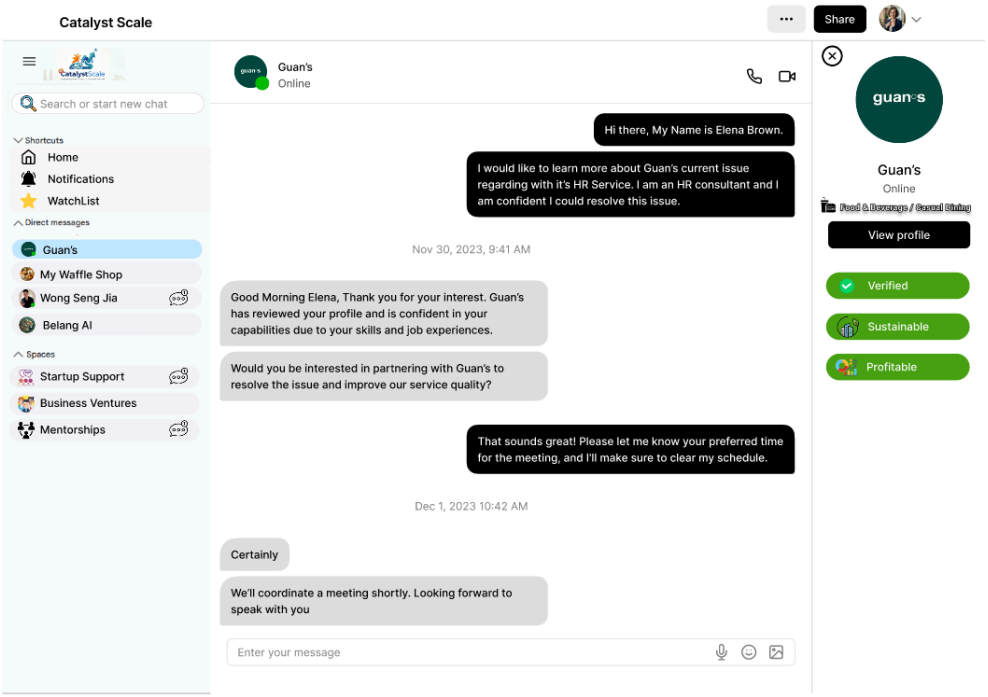
There’s a navigation bar below the large profile icon “Chloe Kelly”, you can choose to edit your investment or professional profile. Tap the <- arrow button on the top left corner to back to the home page.



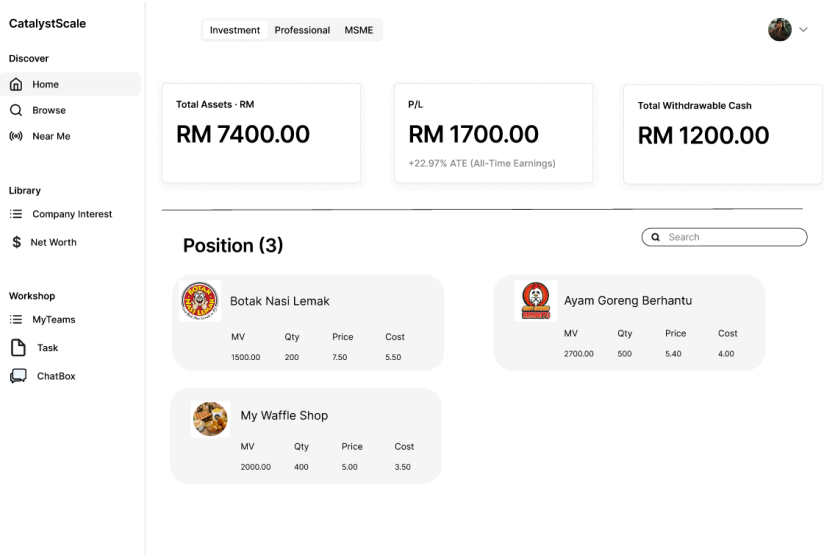
This is the professional profile page, you can choose to edit your skills and job experience



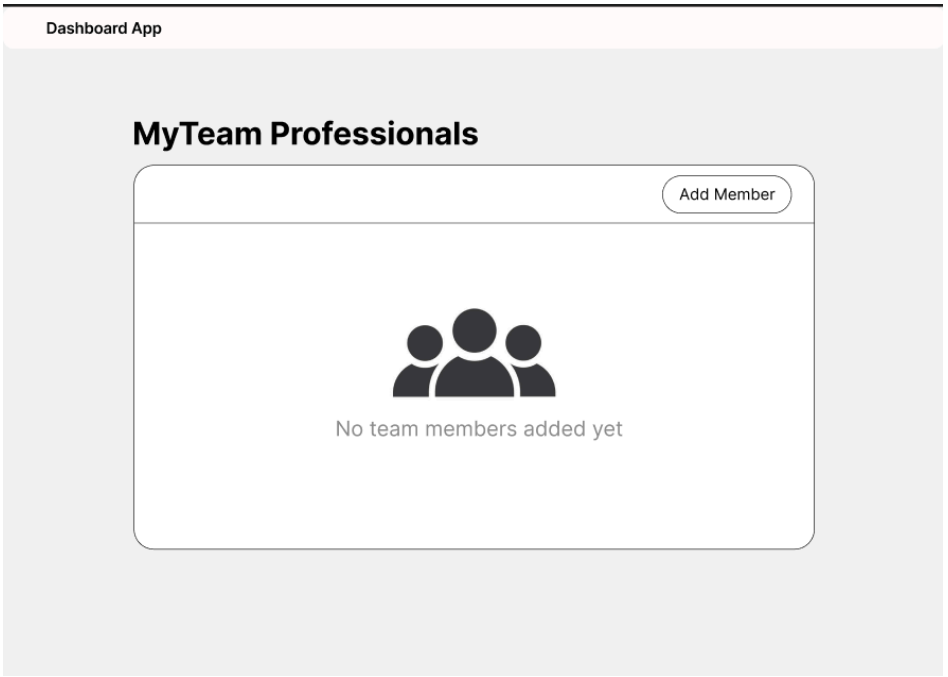
Now, you can tap on the ChatBox icon at the left navigation bar under the ‘Workshop’ to start chatting with companies.



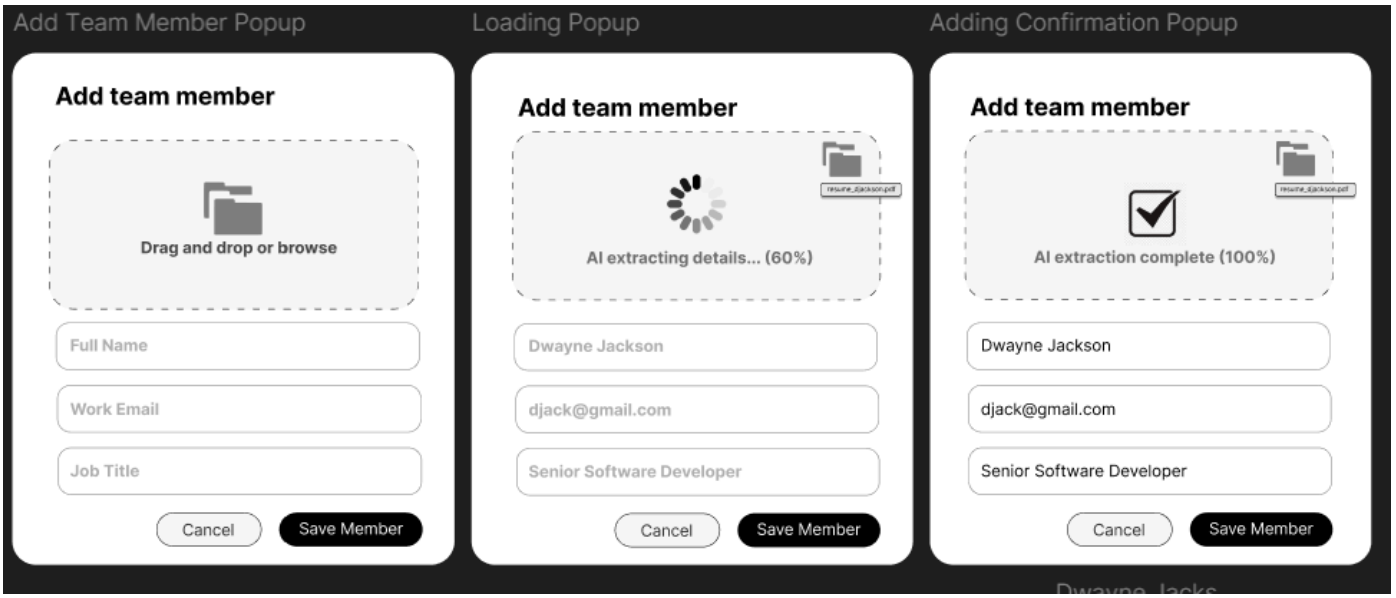
This is the chatbox page, if you want to go back to the home page, just tap on the ‘Home’ icon at the left navigation bar.



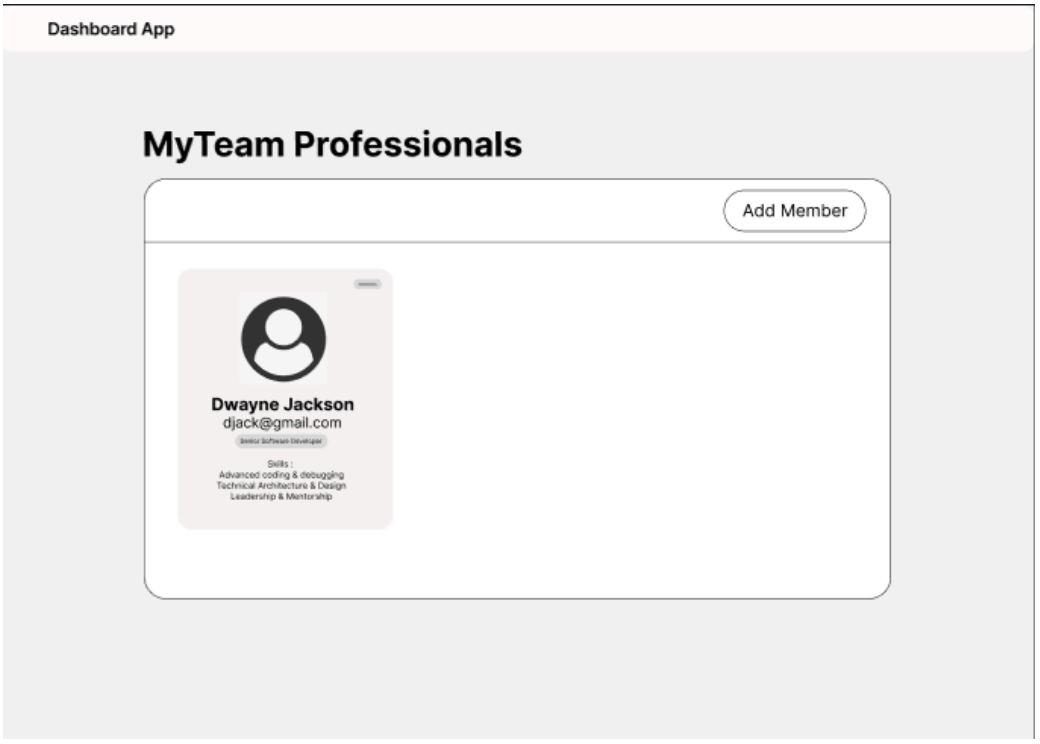
Then, tap on the MyTeams icon at the left navigation bar under the ‘Workshop’ Category to process to the MyTeam Professional page.



In this page, you can choose view member and add members by tapping on the ‘Add Member’ icon at the top right corner of the dashboard.

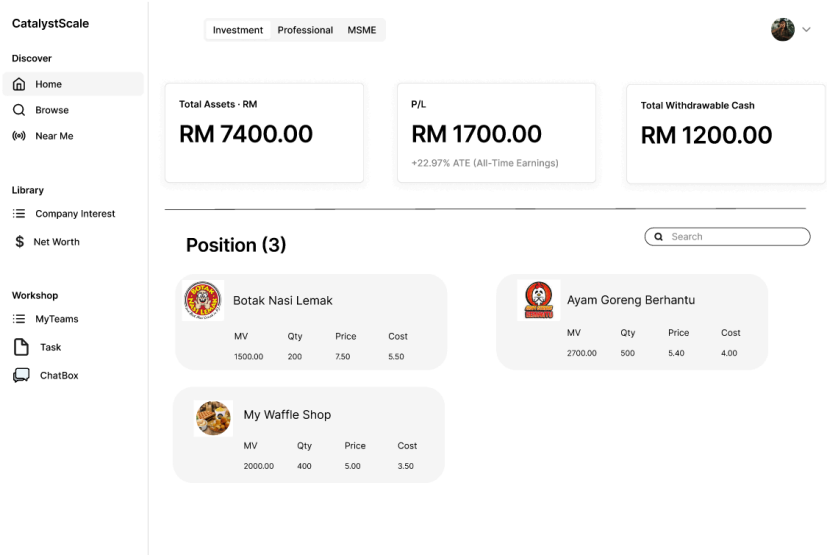


This is the popup page to add team members. You will need to either drag and drop the resume files or browse from your device. Then AI will extract the main details of the resume and then automatically fill up the main information below, you can choose to edit it afterwards. Then tap on save member and you have successfully added a new member.

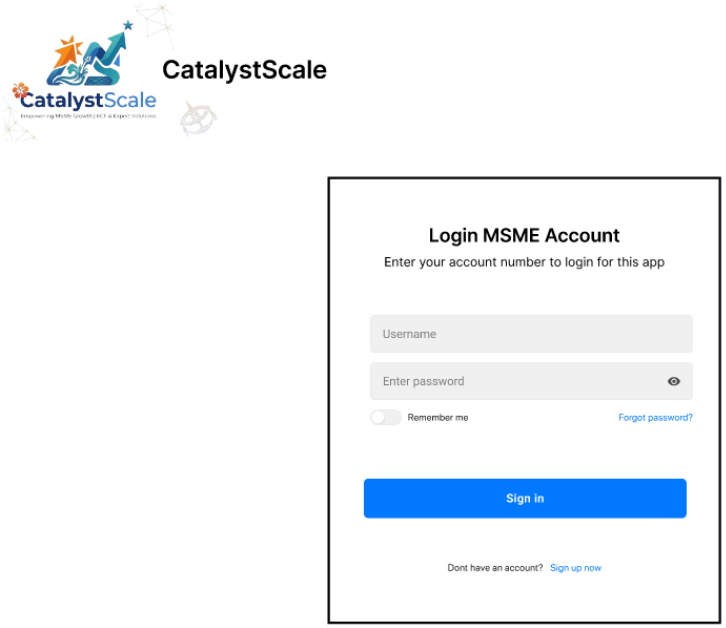


This is the page after you have successfully added a new member. You can tap on the profile icon on each individual to view his/her detail.

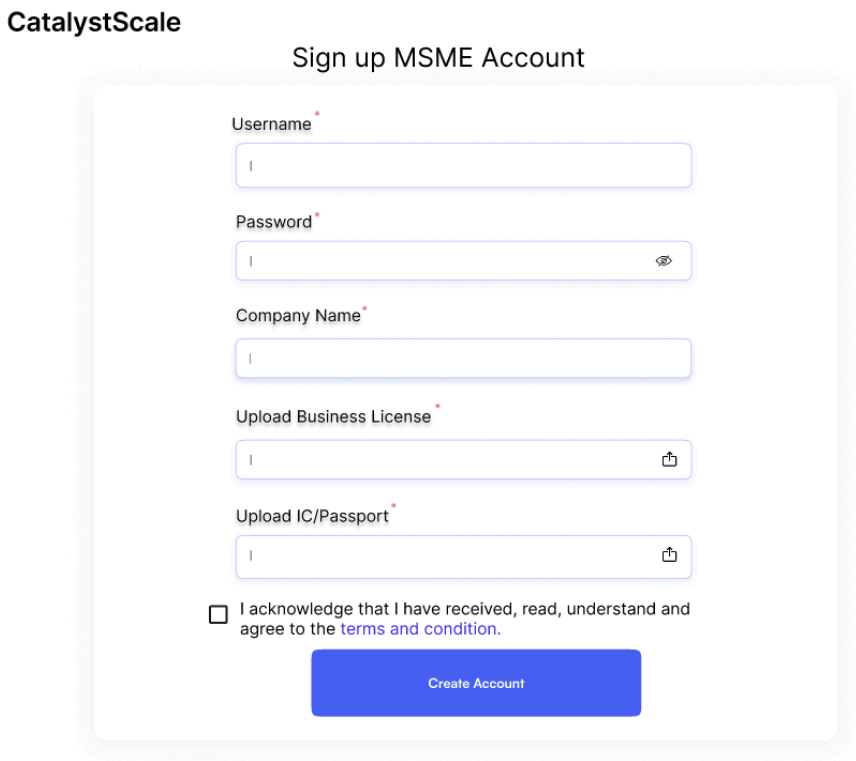
Flow 2: MSME Account



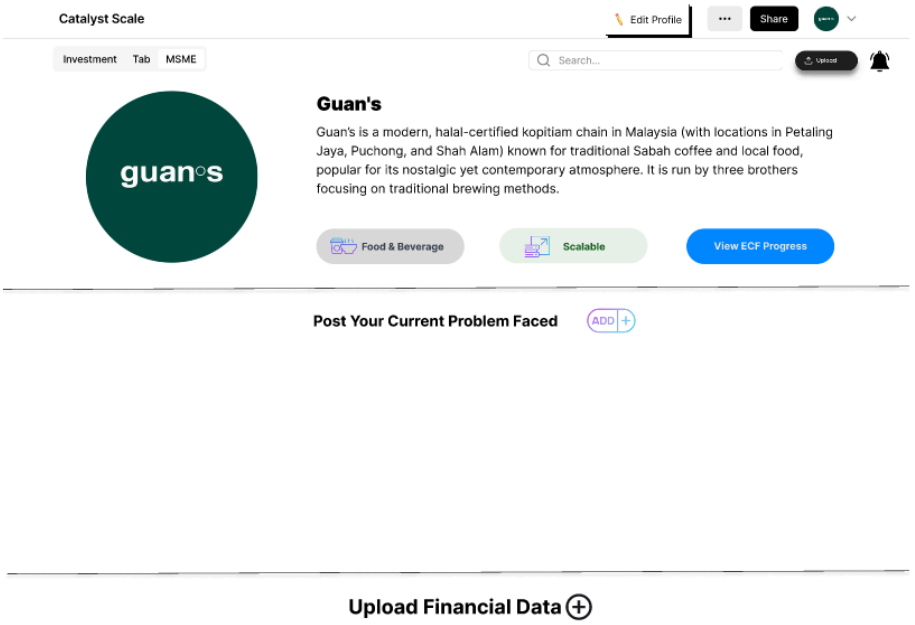
Now, tap on the ‘MSME’ at the navigation bar above beside ‘Professional’ to proceed to the MSME page.



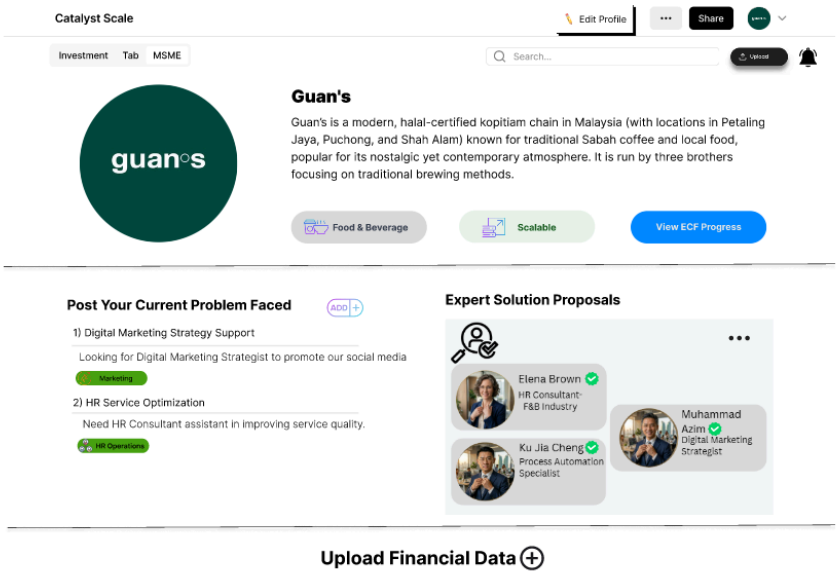
You will need to log in to your MSME account if you haven’t do so, tap on the sign up now button if you don’t have a MSME account.



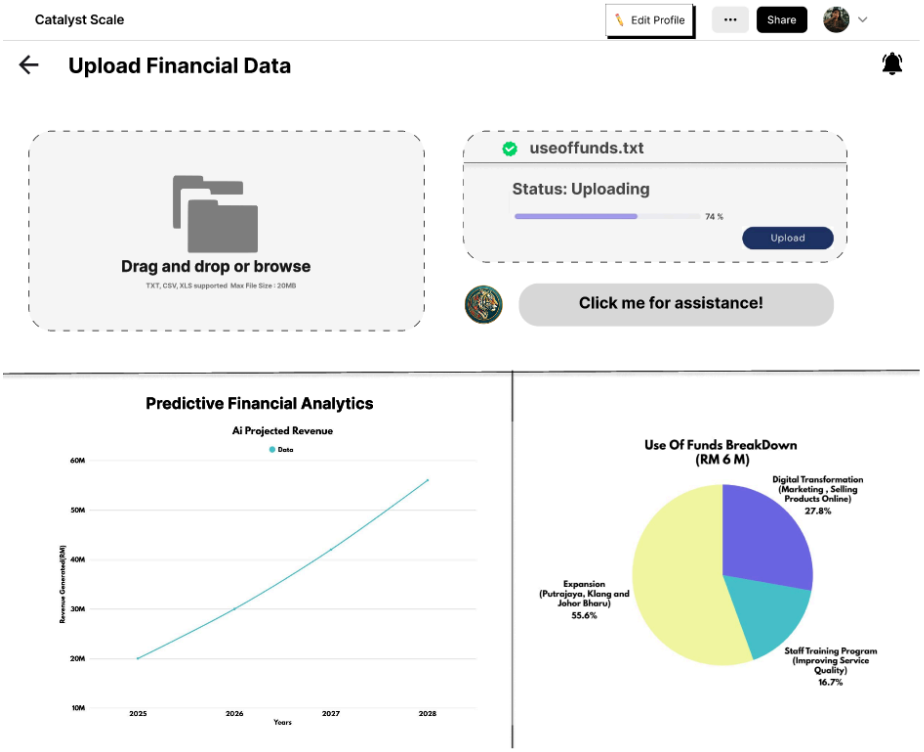
Fill in the details and tap on the create account button to create your MSME account.



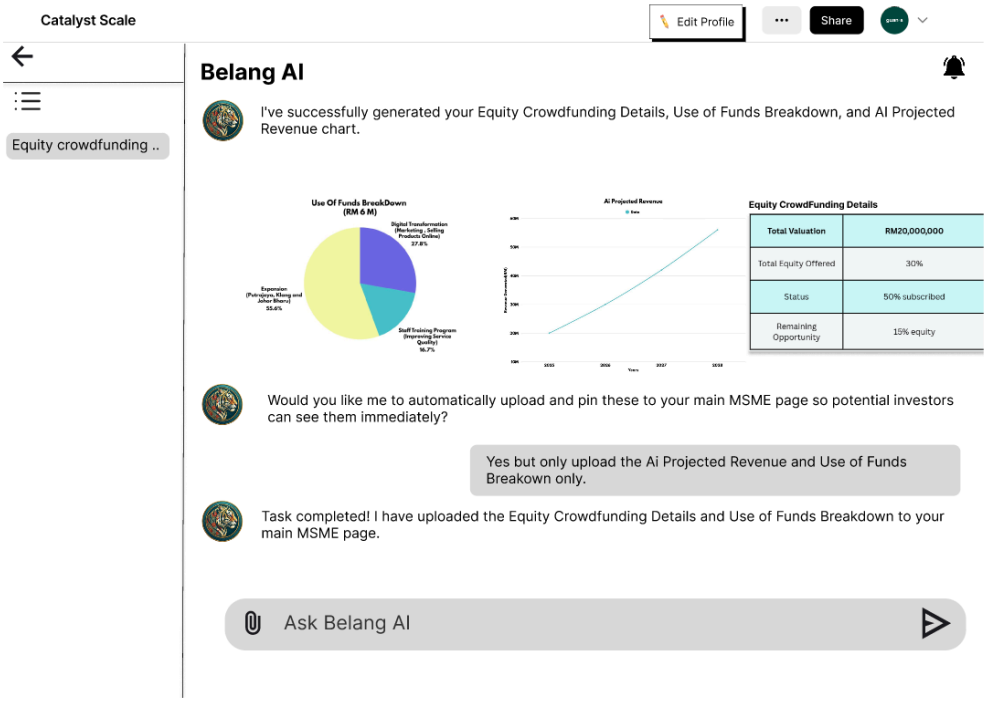
Tap on the add button to post your current problems faced



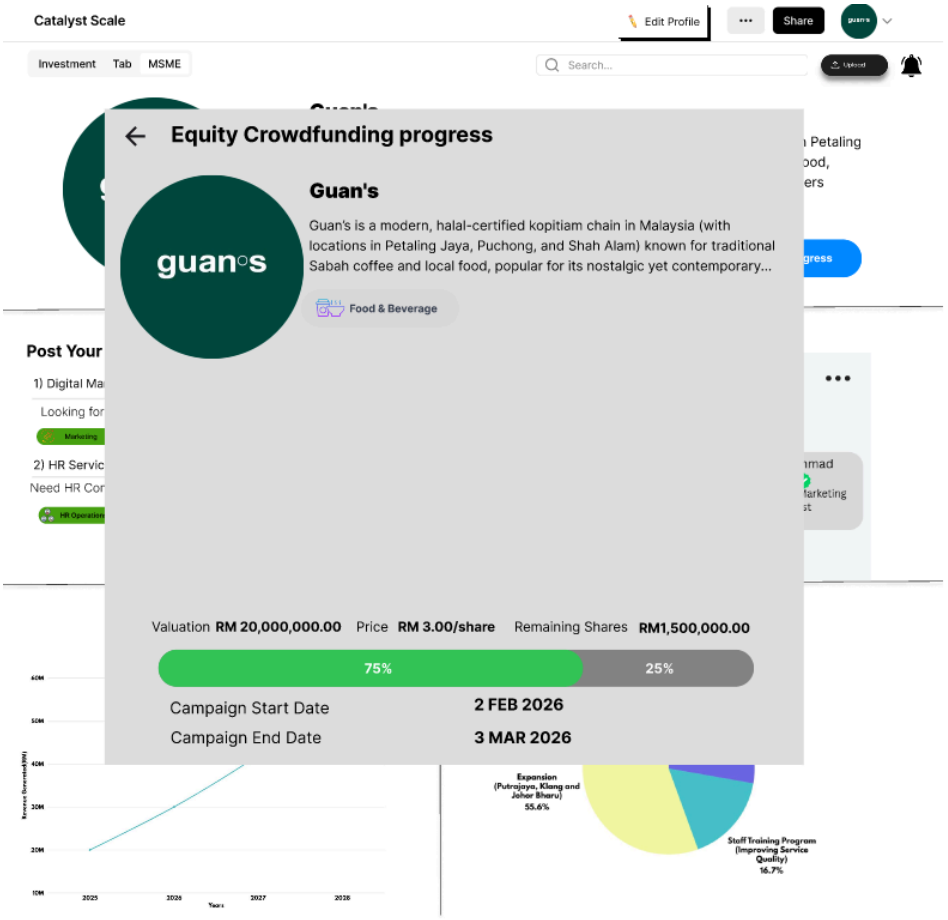
Then, tap on the + button to upload financial data.



Upload your financial data, then click on the ‘Click me for assistance’ to let Belang AI help you generate your equity crowdfunding details.

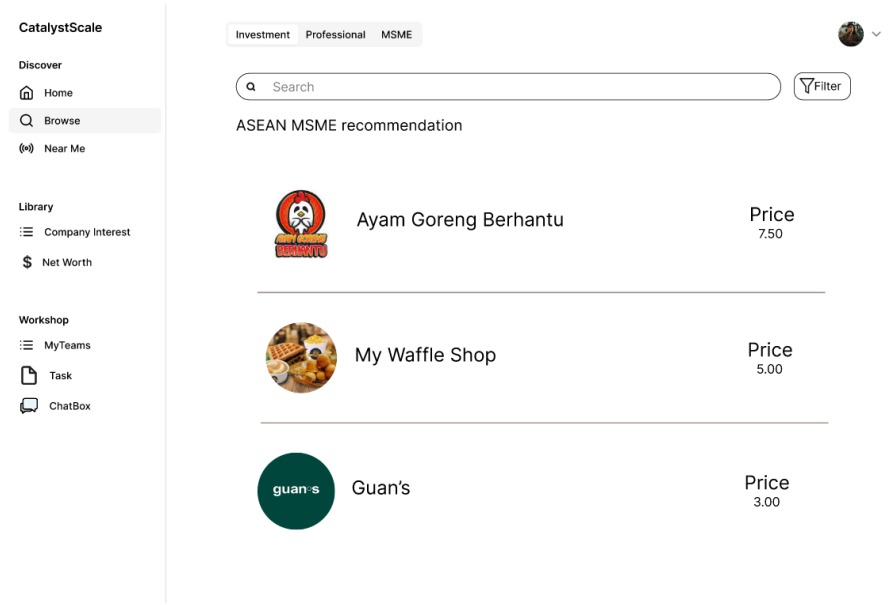


You can chat with the Belang AI regarding the uploads and updates, it will help you solve the problems.

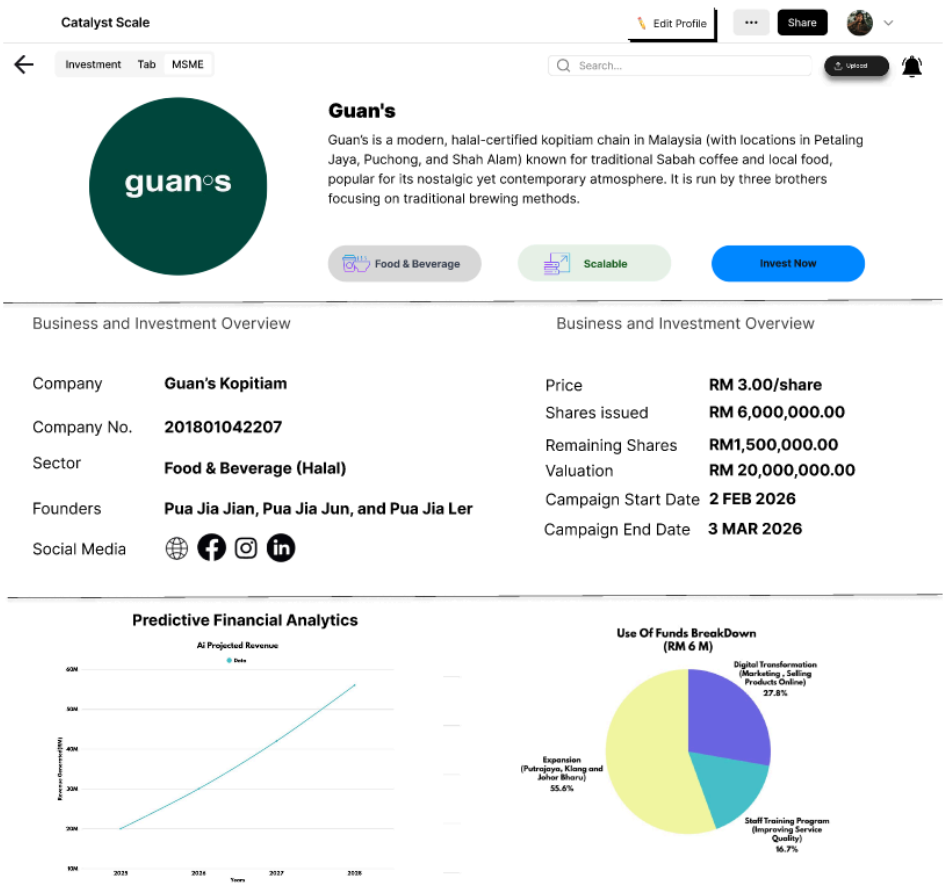


Then, tap on the ECF progress to view your equity crowdfunding progress.

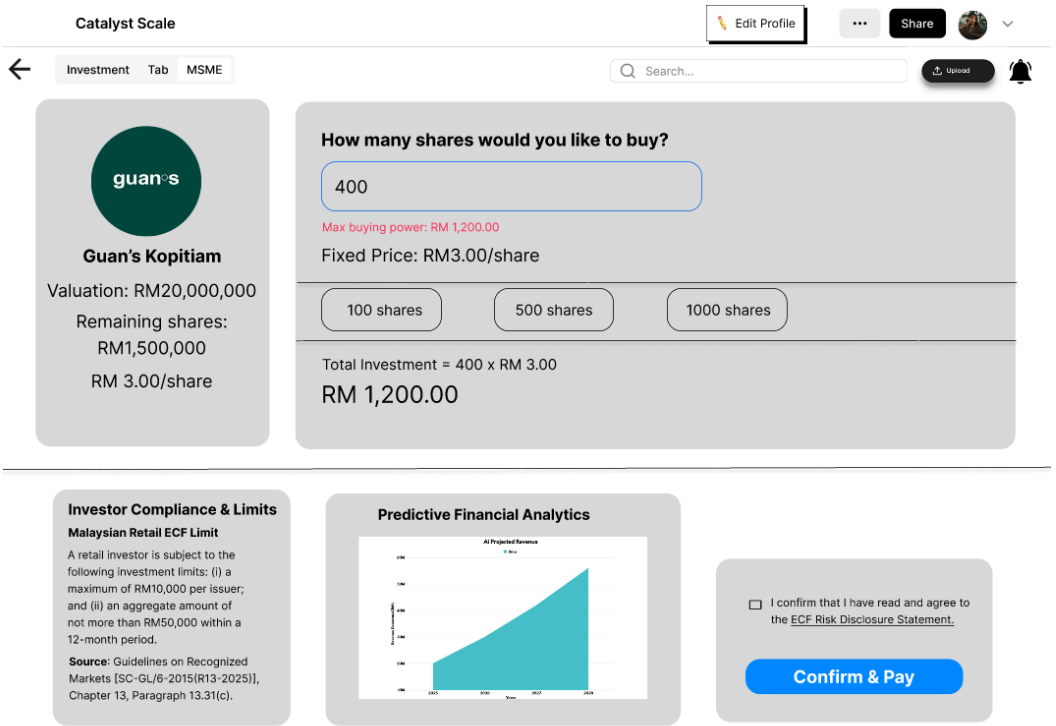
Flow 3: Investing Procedure (Investors view)



Tap on the Browse button on the left navigation bar under 'Discover' category to look for MSME you want to invest in. Then tap on the company to view its details.



This is the company details page. Tap on the blue ‘Invest Now’ button to start investing.



Then key in the amount of shares you like to buy and then tap on the blue ‘Confirm & Pay button’ to start investing.

AI Acknowledgement

Disclosure of AI Tools:

- **Gemini (Google):** Primary AI collaborator for research and documentation.

Usage:

- **Strategic Planning:** Used **Gemini** to structure the design thinking process, refine the business logic for the MSME-Investor matching engine, and draft the competitive market analysis.

Conclusion

Project Summary Our project addresses the critical "Inclusion Gap" facing MSMEs by providing an AI-driven ecosystem that transforms raw business data into growth opportunities. By allowing business owners to upload financial records for AI diagnosis, connect with professional advisors via a collaborative hub, and gain visibility with verified investors, we have created a "Growth Loop" that democratizes success. Our Figma prototype demonstrates that high-level business strategy and investment readiness are no longer reserved for large corporations, but can be made accessible to any entrepreneur with a smartphone.

Future Improvements To scale this project beyond the hackathon, we have identified three key areas for development:

1. **Localized Language Models (LLMs):** Implementing support for regional dialects (such as Bahasa Melayu and local Bornean dialects) via voice-to-text to ensure even lower barriers for non-tech-savvy users.
2. **Blockchain-Verified Credit Scoring:** Integrating blockchain technology to create an immutable record of an MSME's financial history, providing investors with even higher levels of trust and security.
3. **Automated Regulatory Compliance:** Expanding the AI's capabilities to help MSMEs automatically generate the necessary paperwork for government grants and tax filings, further reducing administrative burdens.